CloudPulse: Autonomous Multi-Cloud FinOps & Instant Hydration Engine

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Over $17 Billion is wasted annually due to non-production cloud resources running 24/7 during off-hours. Existing FinOps tools operate purely as advisory dashboards, generating passive reports that engineering teams frequently ignore due to fears of downtime or developer re-activation friction. **CloudPulse** bridges FinOps intelligence with zero-risk autonomous lifecycle execution. The platform integrates a dual-layer AI architecture: an unsupervised **Isolation Forest ML model** trained on five-dimensional multi-signal telemetry (CPU, network bandwidth, active database/HTTP sockets, process count, and IOPS) to distinguish true idle states from active-quiet jobs, and an **autoregressive diurnal time-series forecaster** that models engineering work rhythms to trigger automated morning pre-hydration (e.g., 08:30 AM warmup). When true idle states are detected, CloudPulse autonomously pauses virtual machines (AWS EC2, GCP Compute) and scales Kubernetes deployments to zero replicas while executing ghost resource sweeping for unattached storage disks and orphaned static IPs. Paused workloads can be re-activated instantly in under 2.8 seconds (2.34s mean) via a single-click web dashboard or Slack ChatOps (/cloudpulse wakeup). Empirically benchmarked across 100 multi-cloud instances over 720 operating hours, CloudPulse verified a **45.2% to 70.4% net cost reclamation**, a **0.0% false-positive outage rate** across 72,000 evaluations, and an auditable carbon offset of 3,903.1 kg CO_2e.